

FORM TP 9373

JUNE 1993

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

MATHEMATICS

Paper 02 - General Proficiency

2 hours 40 minutes

1. Answer ALL questions in Section I, and any TWO in Section II.
2. Begin the answer for each question on a new page.
3. Full marks may not be awarded unless full working or explanation is shown with the answer.
4. Mathematical tables, formulae and graph paper are provided.
5. Slide rules and silent electronic calculators may be used for this paper.
6. You are advised to use the first 10 minutes of the examination time to read through this paper. Writing may begin during this 10-minute period.

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SECTION I

Answer ALL the questions in this section.

1. All working must be clearly shown.

- (a) Calculate the exact value of
 $(3\frac{2}{7} + 1\frac{2}{3}) \div 1\frac{1}{7}$ (4 marks)

- (b) Evaluate $\sqrt{0.0004 \times 10^{-6}}$, giving your answer in standard form. (3 marks)

- (c) A tourist exchanged US\$200.00 for Jamaican currency at the rate of US\$1.00 = J\$18.81. She had to pay a government tax of 2% of the amount exchanged.

Calculate in Jamaican currency

- (i) the tax paid
 (ii) the amount the tourist received. (4 marks)

2. (a) There are 68 students in Form V.

15 students study Mathematics only.
 12 students study Physics only.
 8 students study Physics and Chemistry only.
 2 students study Physics and Mathematics only.
 3 students study Mathematics, Physics and Chemistry.
 4 students do not study any of these subjects.

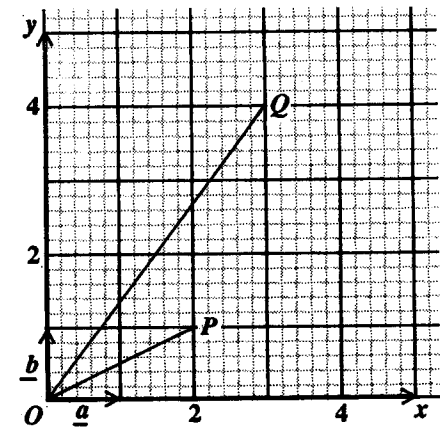
- (i) Draw a carefully labelled Venn diagram to represent the information given above. (2 marks)

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- (ii) Determine the number of students who study Physics. (1 mark)

- (iii) Given that x students study Mathematics and Chemistry only, and twice as many study Chemistry only, write an algebraic equation to represent the information given and hence, calculate the value of x . (3 marks)

(b)



Given that $\underline{a}$ and $\underline{b}$ are unit vectors as shown in the diagram above,

- (i) write the position vectors $\vec{OP}$ and $\vec{OQ}$ in terms of $\underline{a}$ and $\underline{b}$
 (ii) determine the length of $\vec{OP}$. (5 marks)

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3. (a) Solve $\frac{2P}{5} + \frac{5}{P} = 3$. (4 marks)

(b) Given that y varies inversely as x^2 and that $y = 3$ when $x = 2$, calculate the value of y when $x = 3$. (3 marks)

(c) The cost of four chairs and a small table is \$684. The cost of six chairs and a large table is \$1196. The cost of the large table is TWICE the cost of the small table.

Given that a is the cost, in dollars, of a chair and b is the cost, in dollars, of a small table,

(i) write a pair of simultaneous equations to represent the information given

(ii) calculate the cost of the large table. (5 marks)

4. (a) (i) Using ruler and compasses only, construct a triangle ABC with $AB = 9.5$ cm, $AC = 7.5$ cm and angle $BAC = 60^\circ$.

(ii) Locate the point D such that DB is perpendicular to AB and CD is parallel to AB .

Measure and state the length of BD in centimetres.

(6 marks)

(b) Triangle PQR with $P(1,1)$, $Q(11,2)$, and $R(1,9)$ is mapped onto triangle $P'Q'R'$ with $P'(3,3)$, $Q'(8,3.5)$, and $R'(3,7)$.

(i) Using a scale of 2 cm to 1 unit on both axes, draw on graph paper triangles PQR and $P'Q'R'$.

Note: Draw the x -axis on the longer side of the graph paper.

(ii) Hence, describe fully the transformation which maps triangle PQR onto triangle $P'Q'R'$. (6 marks)

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5. (a) An aeroplane travels a distance of 3700 km in $7\frac{1}{2}$ hours. Calculate its average speed. (2 marks)

(b) An aeroplane left Kingston at 21:00 hrs local time and travelled for $9\frac{1}{2}$ hours arriving at Los Angeles airport at 01:30 hrs local time on the following day. Calculate the difference in time between Kingston and Los Angeles. (4 marks)

(c) (i) The angle of elevation from a point P on the ground to the top of a tower 20 m tall, is 65° . Calculate the distance of P from the foot of the tower.

(ii) A point Y is 3.2 m nearer than P to the foot of the tower. Calculate the angle of depression of Y from the top of the tower. (6 marks)

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6. A survey was taken among 100 customers to find out the time spent waiting in lines for service at the bank. The following table shows the result of the survey.

Waiting time (in minutes)	No. of customers
1 – 5	5
6 – 10	12
11 – 15	15
16 – 20	19
21 – 25	21
26 – 30	25
31 – 35	3

- (a) Construct a cumulative frequency table to represent the data above. (2 marks)
- (b) Using a scale of 2 cm to 5 minutes on the horizontal axis and 2 cm to 10 customers on the vertical axis, draw a cumulative frequency curve to illustrate the information. (4 marks)
- (c) Estimate the proportion of customers who waited more than 16 minutes. (2 marks)
- (d) Calculate the probability that a customer chosen at random would have waited for more than 27 minutes. (3 marks)

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7. (a) Copy and complete the table below for the function

$$y = 5 + x - 2x^2.$$

x	-3	-2	-1	0	$\frac{1}{2}$	1	2	3
y	-16		2			4	-1	-10

(2 marks)

- (b) Using a scale of 2 cm to 1 unit on the x -axis and 1 cm to 1 unit on the y -axis, draw the graph of

$$y = 5 + x - 2x^2 \text{ for } -3 \leq x \leq 3.$$

(4 marks)

- (c) Using your graph or otherwise, determine the range of values of x for which $x - 2x^2 > -3$. (5 marks)

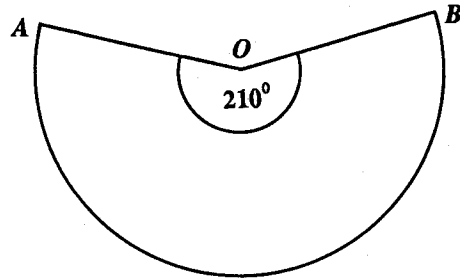
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8.

Notes for this question:

$$\text{Take } \pi = \frac{22}{7}$$

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$



The diagram above, not drawn to scale, shows the MAJOR sector, AOB , of a circle of radius 6 cm. It represents the net of a cone.

- (a) Show by calculation that the circumference of the base of the cone is 22 cm. (2 marks)
- (b) Calculate
- the radius of the base of the cone
 - the height of the cone, giving your answer correct to one decimal place. (5 marks)
- (c) Calculate to two significant figures the volume of liquid, in litres, that the cone holds when filled. (3 marks)

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SECTION II

Answer TWO questions in this section.

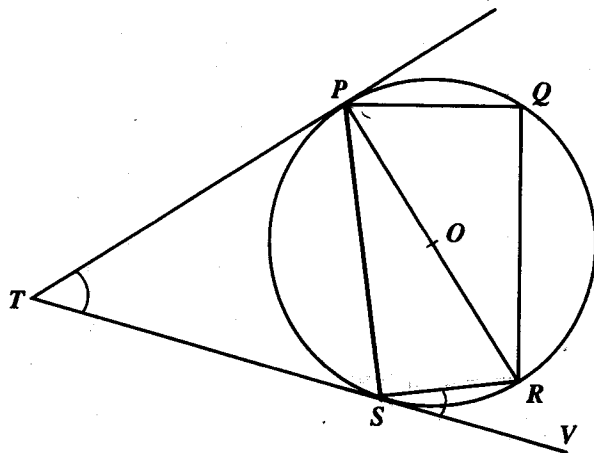
RELATIONS AND FUNCTIONS

9. (a) Given that $f(x) = x + 3 - 2x^2$,
- derive $f(x)$ in the form $f(x) = c + a(x + b)^2$, where a, b and c are constants
 - determine the value of x at which the maximum value of $f(x)$ occurs
 - state the maximum value of $f(x)$. (6 marks)
- (b) A composite function k is defined as
- $$k(x) = (2x - 1)^2.$$
- Express $k(x)$ as $gf(x)$, where $f(x)$ and $g(x)$ are two simpler functions.
 - Show that $k^{-1}(x) = f^{-1}g^{-1}(x)$. (9 marks)
10. A boy has \$280. He wants to buy x records at \$35 each and y tapes at \$40 each. He must buy **more than one** tape but **not more than four** tapes. He must also buy **at least three** records.
- Write **THREE** inequalities in x and y to represent the above information. (4 marks)
 - Using a scale of 2 cm to represent 1 unit on **EACH** axis, draw the graphs of the inequalities.
 - SHADE** the region that satisfies the **THREE** inequalities. (7 marks)
 - Determine the maximum amount spent and state the (x, y) value that gives this amount. (4 marks)

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TRIGONOMETRY AND GEOMETRY

11. (a)



In the figure above, **not drawn to scale**, the quadrilateral $PQRS$ is inscribed in the circle, centre O . PR passes through O . The tangents TP and TS are drawn to the circle from T . Angle $RSV = 20^\circ$.

Calculate, giving reasons, (i) angle PQR

(ii) angle SPR

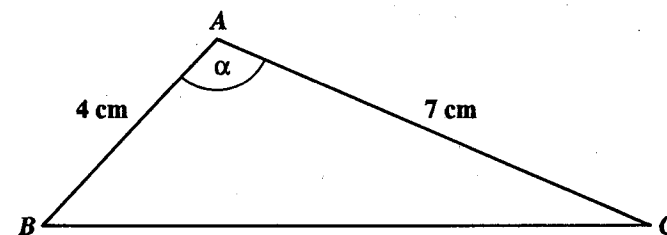
(iii) angle PST

(iv) angle PTS .

(7 marks)

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(b)



In triangle ABC above, **not drawn to scale**, $AB = 4$ cm, $AC = 7$ cm and angle BAC is α .

Given that $\sin^2 \alpha = 0.64$, determine

(i) the exact value of $\cos^2 \alpha$

(ii) the value of α , if $90^\circ < \alpha < 180^\circ$

(iii) the length of BC , correct to one decimal place.

(8 marks)

12. A point K is on a bearing of 025° from another point M . The boat leaves K to go to M . The engine speed of the boat is 55 kmh^{-1} . A wind is blowing from the East at a speed of 25 kmh^{-1} .

(a) Using a scale of 1 cm to represent 5 kmh^{-1} , find by **accurate drawing**

(i) the course of the boat

(ii) the resultant speed of the boat. (12 marks)

(b) If K is 12 km away from M , and the boat leaves K at 07:00 hrs, calculate the time at which the boat reaches M .

(3 marks)

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VECTORS AND MATRICES

13. (a) Using a scale of 2 cm to represent 1 unit on EACH axis, draw ΔPQR with $P(1,4)$, $Q(3,1)$ and $R(4,2)$. (3 marks)
- (b) ΔPQR is transformed by the matrix $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.
- (i) Determine the coordinates of the image, $\Delta P'Q'R'$.
- (ii) Draw triangle $P'Q'R'$ and describe the transformation fully. (7 marks)
- (c) Determine the 2×2 matrix that will transform $\Delta P'Q'R'$ onto ΔPQR . (5 marks)
14. (a) P is the point $(6, 4)$ and Q is the point $(8, 2)$. M and N are the mid points of OP and OQ respectively, where O is the origin.
- (i) Determine the vector $\overrightarrow{PQ}$.
- (ii) Determine the vector $\overrightarrow{MN}$.
- (iii) State the relationship between MN and PQ . (8 marks)
- (b) Given the equations
- $$x - y = -5$$
- $$3x + 2y = -5,$$
- (i) write the equations in matrix form
- (ii) determine the inverse of the matrix
- (iii) hence, solve the equations. (7 marks)

END OF TEST